1. What are the 3 main connectivity modes in Power BI?

The three main connectivity modes in Power BI are Import, DirectQuery, and Live Connection. Import loads data into Power BI for faster analysis, DirectQuery keeps data in the original source and queries it when needed, and Live Connection connects directly to an existing semantic model or Analysis Services model.

2. What are the two gateways in Power BI?

The two gateways are On-premises Data Gateway (Standard Mode) and On-premises Data Gateway (Personal Mode). Standard Mode is mainly used for organizational and shared data sources, while Personal Mode is intended for individual users and supports refresh for Power BI.

3. How does a gateway work?

A gateway acts as a bridge between Power BI cloud services and on-premises data sources. When Power BI needs data from an on-premises database, the gateway securely connects to the local data source and transfers the required data to Power BI.

Power BI Service (Cloud)

↓

Data Gateway

↓

On-Premises Data Source

(SQL Server / Excel / etc.)

4. Difference between Cloud-based Data and On-premises Data

Cloud-based data is stored on internet-based cloud servers and can be accessed through cloud services. On-premises data is stored on an organization's own local servers or computers and usually requires a gateway to connect it with Power BI Service.

5. Steps to Use a Gateway

i. The basic steps are:
ii. Install the On-premises Data Gateway
iii. Sign in with your Power BI account
iv. Register the gateway
v. Add the required data source and provide connection details
vi. Configure users and permissions
vii. Connect the data source to Power BI
viii. Create or publish the report
ix. Configure scheduled refresh if required.